

M5StickC BLE STRIDE Practice Model

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Date Generated: Sun Aug 16 2026



OWASP Threat Dragon

Executive Summary

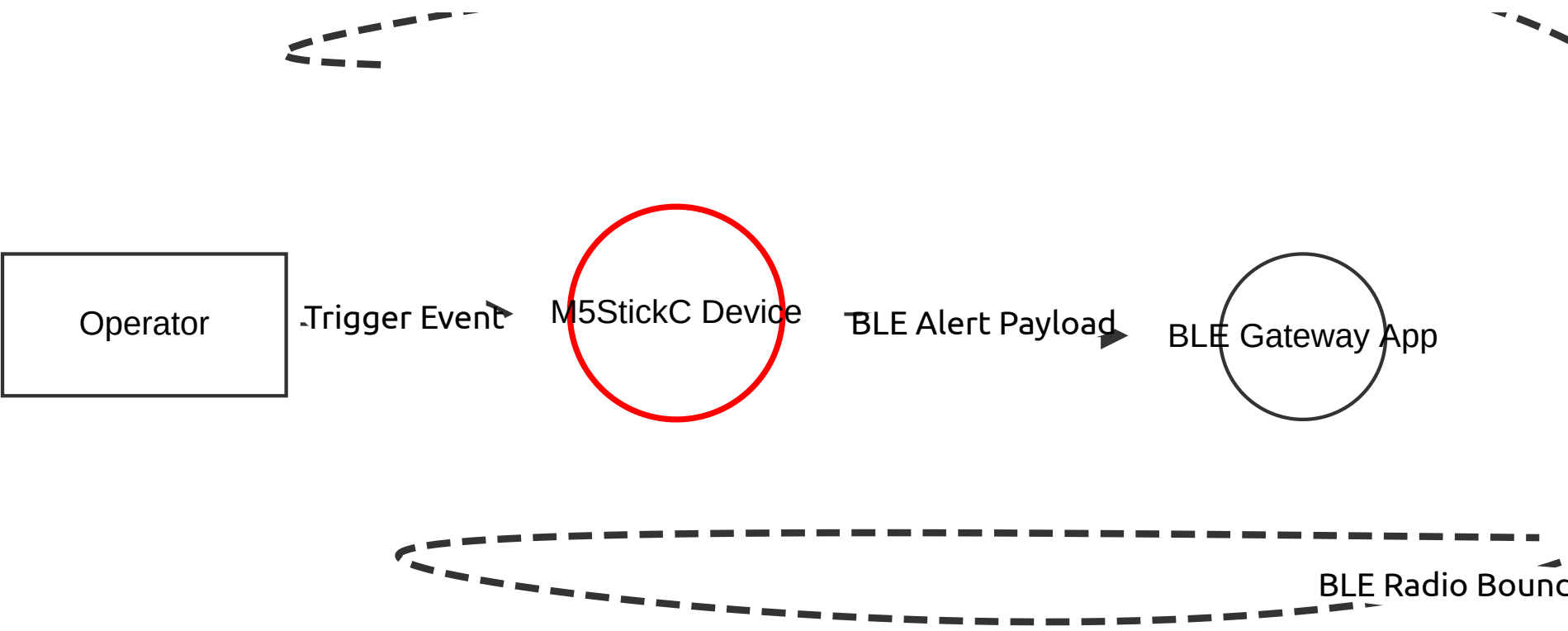
High level system description

Minimal BLE-only STRIDE model for practice: Operator -> M5StickC -> BLE Gateway App

Summary

Total Threats	6
Total Mitigated	0
Total Open	6
Open / Critical Severity	0
Open / High Severity	4
Open / Medium Severity	2
Open / Low Severity	0

BLE STRIDE diagram



BLE STRIDE diagram

Operator (Actor)

Description: Authorized conference operator

Number	Title	Type	Severity	Status	Score	Description	Mitigations
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M5StickC Device (Process)

Description: ESP32 firmware sends BLE alert payload

Number	Title	Type	Severity	Status	Score	Description	Mitigations
1	Spoofed BLE central	Spoofing	High	Open		Unauthorized phone/laptop impersonates trusted BLE gateway and receives device data.	Use bonded pairing, whitelist gateway identity, reject unpaired peers.
2	Tampered BLE payload	Tampering	High	Open		Attacker alters payload/commands over BLE, causing false alert content.	Sign messages (HMAC) with nonce/timestamp and verify before use.
3	No traceable alert origin	Repudiation	Medium	Open		Cannot prove which device/session emitted an alert.	Include device ID and monotonic counter; log timestamped event IDs.
4	BLE metadata disclosure	Information disclosure	Medium	Open		Nearby sniffing reveals alert metadata or identifiers.	Minimize sensitive fields; avoid PII; prefer encrypted BLE characteristics.
5	BLE jamming / disconnect DoS	Denial of service	High	Open		Interference or connection abuse prevents message delivery.	Retry queue with backoff, reconnect limits, and manual fallback path.
6	Privilege escalation via debug/control characteristic	Elevation of privilege	High	Open		Attacker accesses privileged BLE characteristic to trigger admin-only behavior.	Split read/write roles, enforce authz checks per characteristic, disable debug in demo build.

BLE Gateway App (Process)

Description: Receives BLE packet and relays event

Number	Title	Type	Severity	Status	Score	Description	Mitigations
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Trigger Event (Data Flow)

Description: Operator initiates alert

Number	Title	Type	Severity	Status	Score	Description	Mitigations
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BLE Alert Payload (Data Flow)

Description: Single BLE message sent to trusted gateway

Number	Title	Type	Severity	Status	Score	Description	Mitigations
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